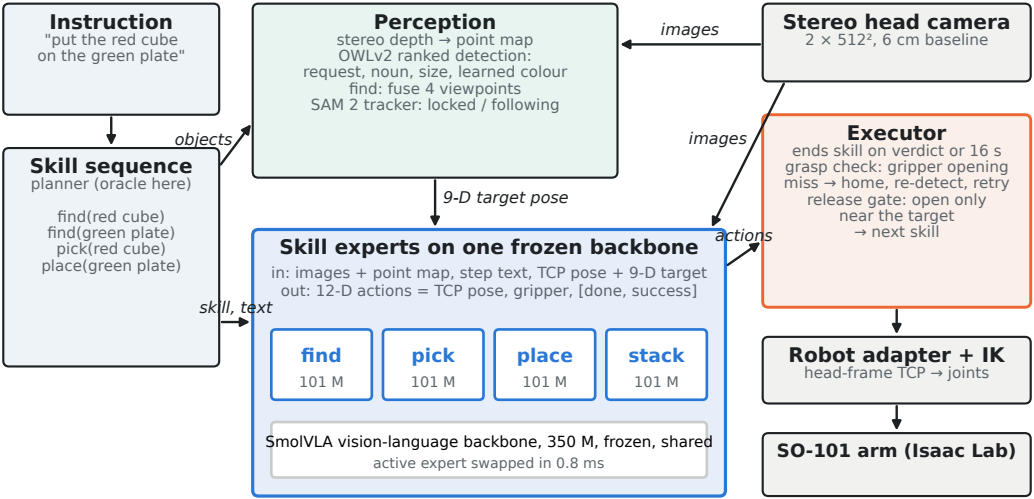
Instruction
"put the red cube on the green plate"
Skill sequence
planner (oracle here)
find(red cube)
find(green plate)
pick(red cube)
place(green plate)
objects
skill, text
Perception
stereo depth → point map
OWLv2 ranked detection:
request, noun, size, learned colour
find: fuse 4 viewpoints
SAM 2 tracker: locked / following
images
Stereo head camera
2 × 512², 6 cm baseline
images
9-D target pose
Executor
ends skill on verdict or 16 s
grasp check: gripper opening
miss → home, re-detect, retry
release gate: open only
near the target
→ next skill
actions
Skill experts on one frozen backbone
in: images + point map, step text, TCP pose + 9-D target
out: 12-D actions = TCP pose, gripper, [done, success]
find
101 M
pick
101 M
place
101 M
stack
101 M
SmolVLA vision-language backbone, 350 M, frozen, shared
active expert swapped in 0.8 ms
Robot adapter + IK
head-frame TCP → joints
SO-101 arm (Isaac Lab)